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Splunk SIEM Deployment & Log Management

Centralized Log Monitoring using Splunk Enterprise and Universal Forwarder

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Professional Portfolio

Project

Splunk Enterprise Deployment, Log Collection, Data Ingestion and Security Monitoring

About this Project

This project demonstrates the implementation of a Security Information and Event Management (SIEM) solution using Splunk Enterprise in a Linux environment.

The deployment included Splunk Enterprise installation, Universal Forwarder configuration, Linux log ingestion, authentication monitoring, event investigation and validation of log collection through Search Processing Language (SPL).

The objective was to simulate real-world Security Operations Center (SOC) activities, providing centralized visibility into infrastructure events while developing practical skills in log management, monitoring and incident investigation.

Key Skills Demonstrated

SIEM

- Splunk Enterprise
- Universal Forwarder
- Data Ingestion
- Index Management
- SPL Queries
- Log Correlation

Security Operations Center (SOC)

- Log Analysis
- Event Monitoring
- Threat Detection Foundations
- Security Investigation
- Incident Analysis

Linux Administration

- Service Management
- User Administration
- Syslog Monitoring
- Auth.log Analysis
- CLI Operations

Project Timeline

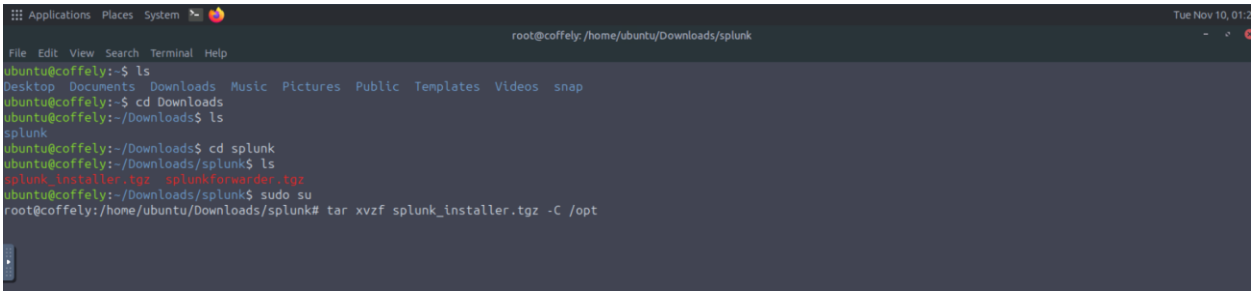
Phase	Status
Splunk Enterprise Installation	✅
Splunk Startup & Configuration	✅
Administrative Account Creation	✅
Service Validation	✅
Universal Forwarder Installation	✅

Port Conflict Resolution	✅
Forwarding & Receiving Configuration	✅
Index Creation (linux_host)	✅
Forward Server Registration	✅
Syslog Monitoring	✅
Data Ingestion Validation	✅
User Activity Monitoring	✅
Security Event Investigation	✅

Featured Screenshots

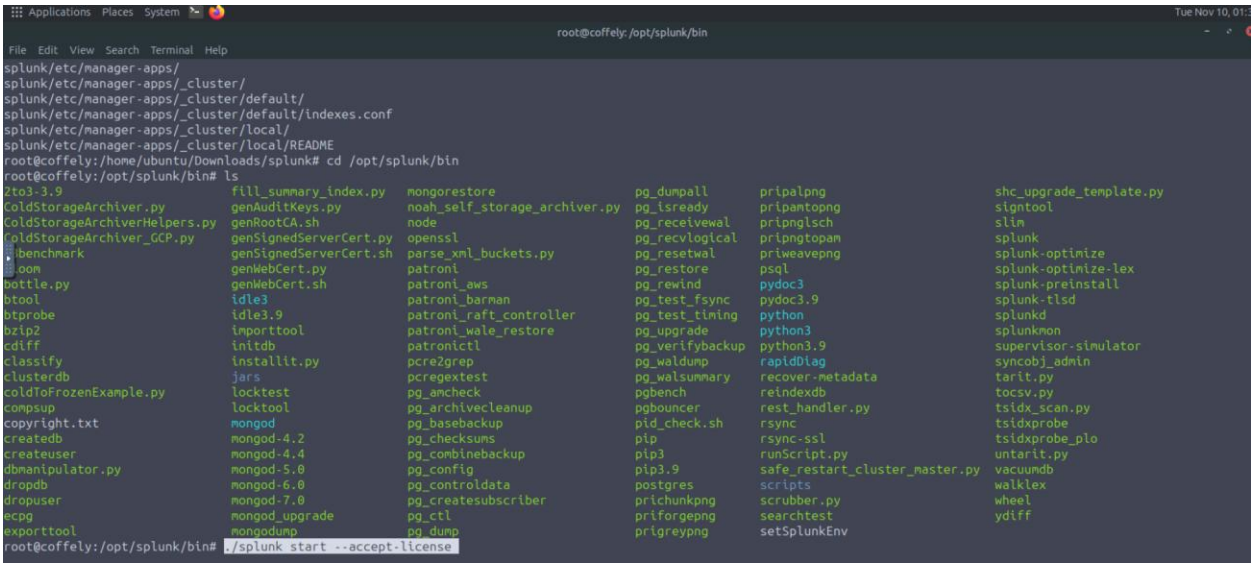
Screenshot 01

Splunk Enterprise Installation



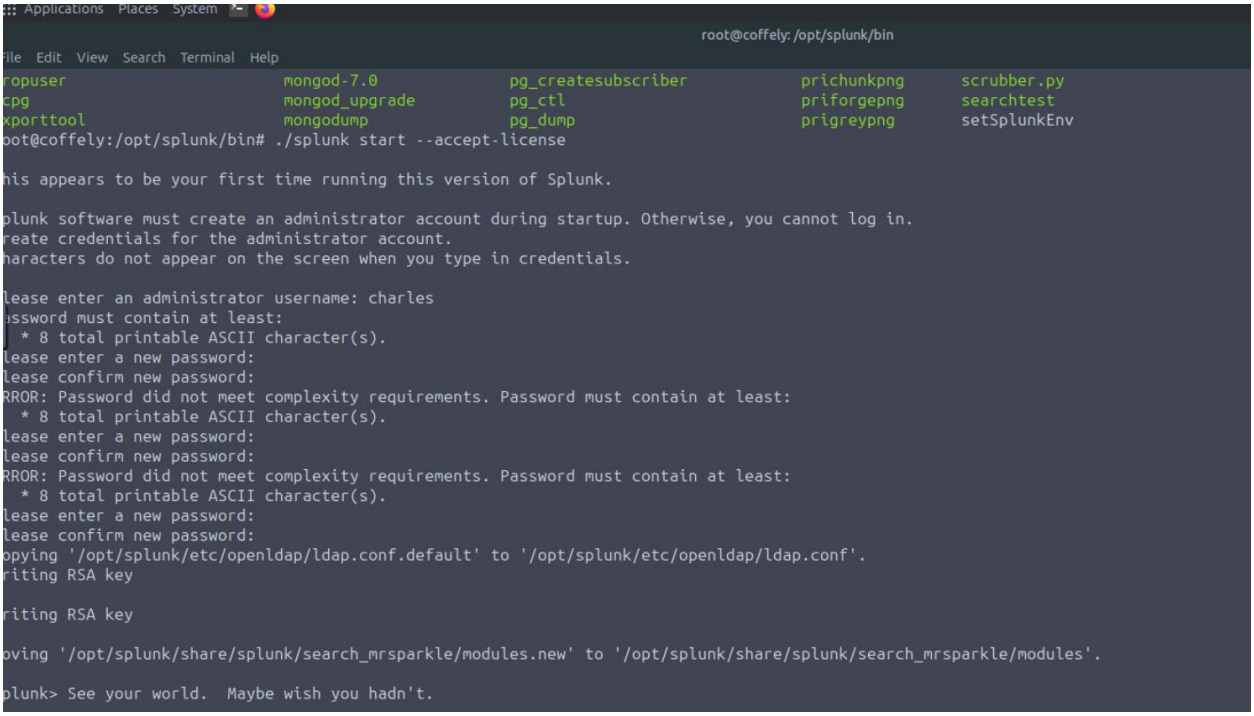
Screenshot 02

Splunk Initial Startup



Screenshot 03

Administrative Account Creation



Screenshot 04

Splunk Status Validation

```
root@coffely:/opt/splunk/bin# ./splunk status
splunkd is running (PID: 3166).
splunk helpers are running (PIDs: 3167 3384 3393 3464 3475).
root@coffely:/opt/splunk/bin#
```

Screenshot 05

Splunk Universal Forwarder Deployment

```
root@coffely:/home/ubuntu/Downloads/splunk# cd /opt/splunkforwarder/bin
root@coffely:/opt/splunkforwarder/bin# ./splunk start --accept-license

This appears to be your first time running this version of Splunk.

Splunk software must create an administrator account during startup. Otherwise, you cannot log in.
Create credentials for the administrator account.
Characters do not appear on the screen when you type in credentials.

Please enter an administrator username: charles
Password must contain at least:
   * 8 total printable ASCII character(s).
Please enter a new password:
```

Screenshot 06

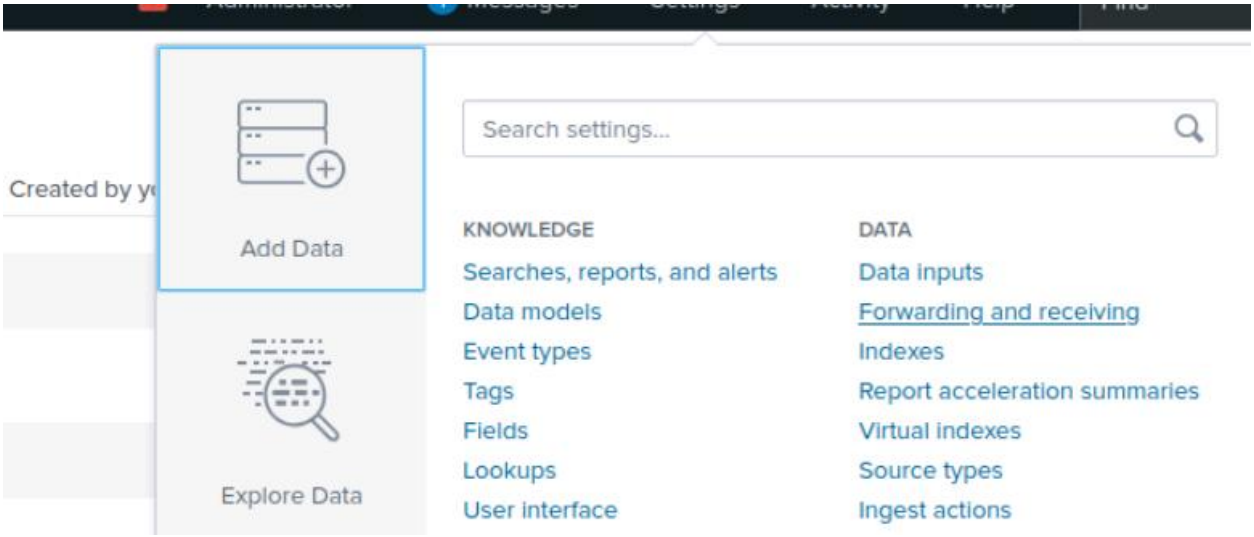
Port Conflict Resolution (8089 → 8090)

```
Splunk> See your world.  Maybe wish you hadn't.

Checking prerequisites...
       Checking mgmt port [8089]: not available
ERROR: mgmt port [8089] - port is already bound.  Splunk needs to use this port.
Would you like to change ports? [y/n]: y
Enter a new mgmt port: 8090
Setting mgmt to port: 8090
The server's splunkd port has been changed.
```

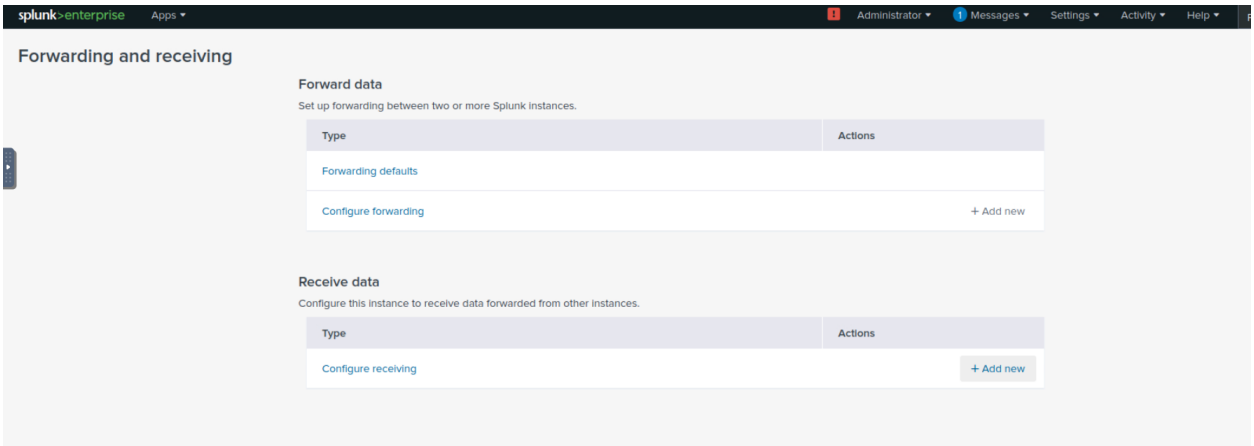
Screenshot 07

Forwarding and Receiving Configuration



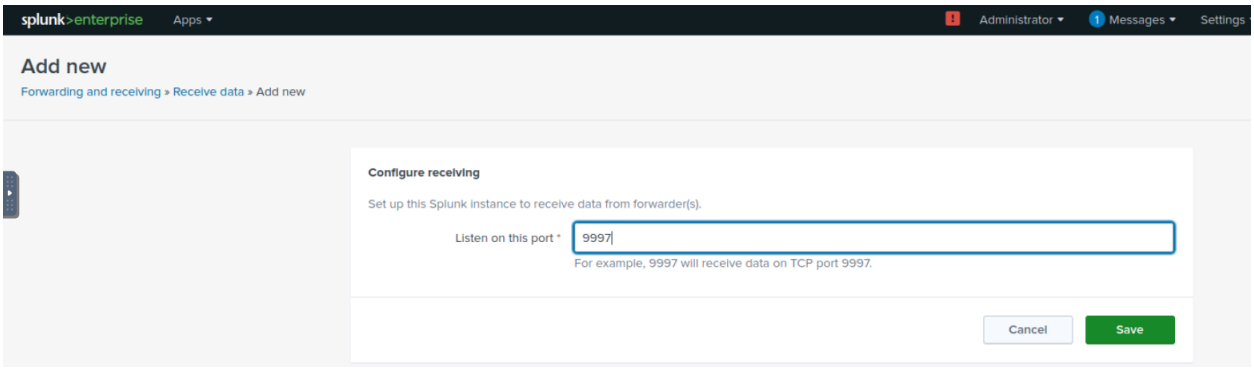
Screenshot 08

Receive Data Configuration



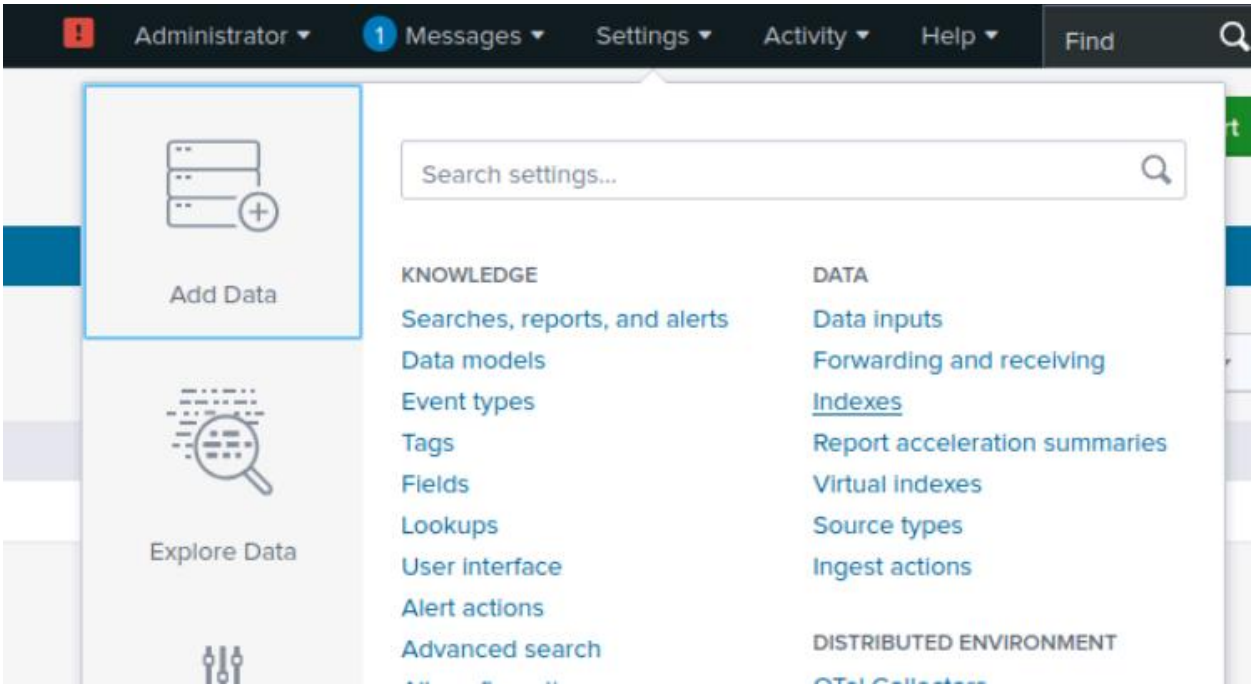
Screenshot 09

TCP Port 9997 Configuration



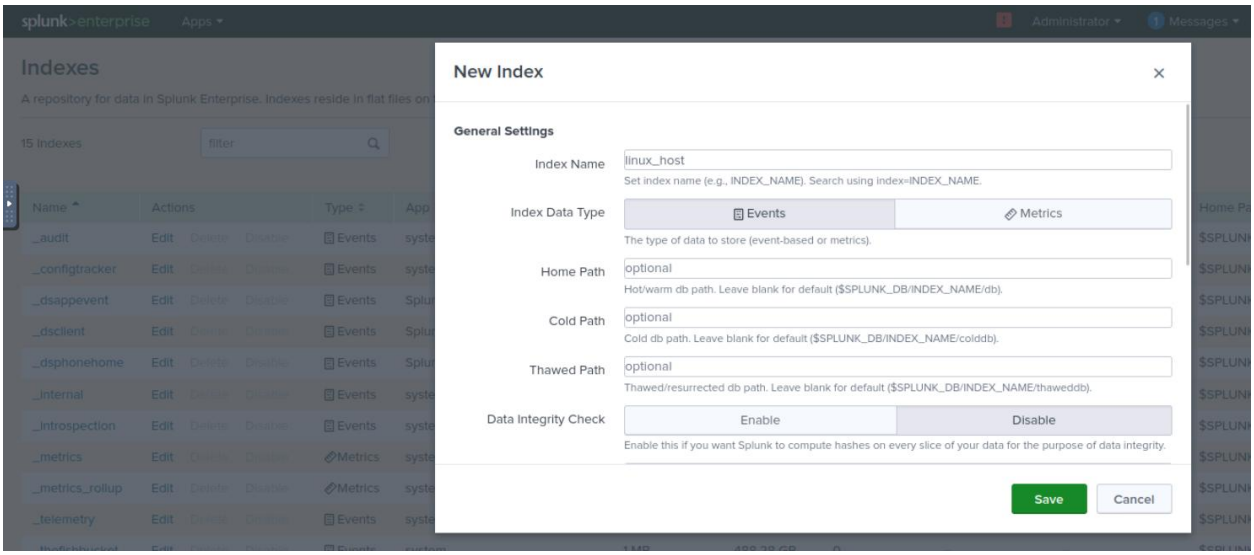
Screenshot 10

Index Management Access



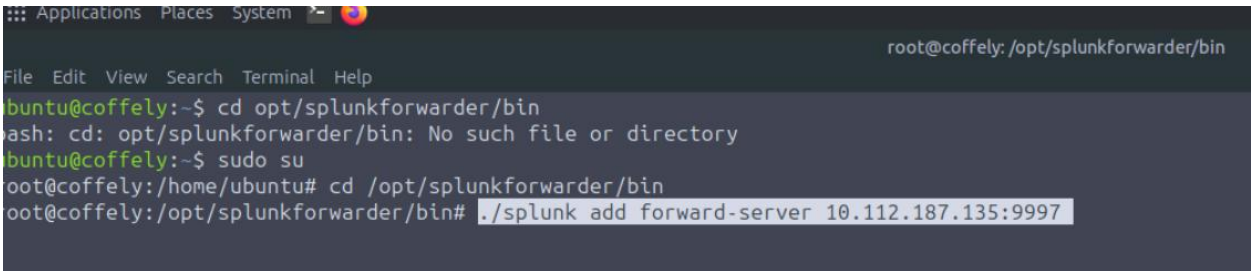
Screenshot 11

linux_host Index Creation



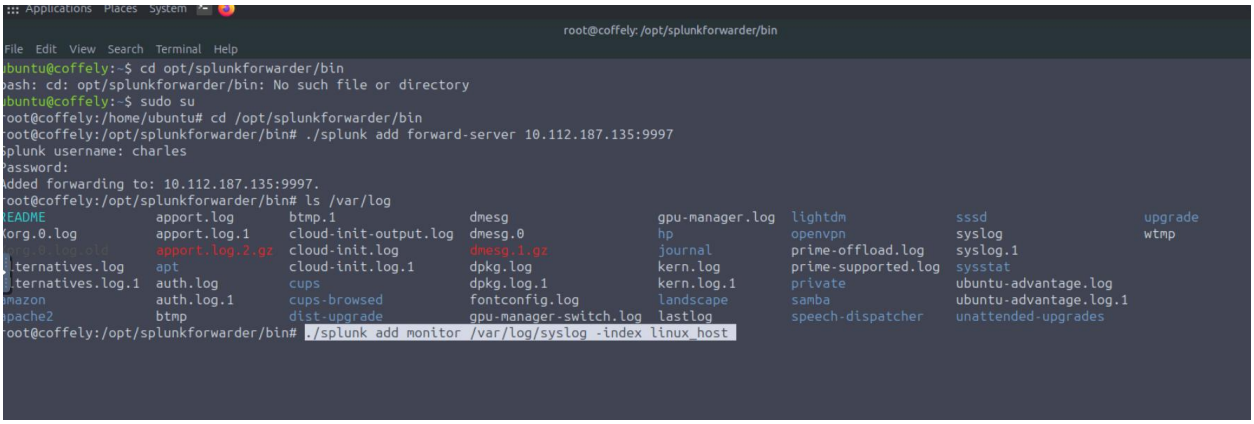
Screenshot 12

Forward Server and Log Source Configuration



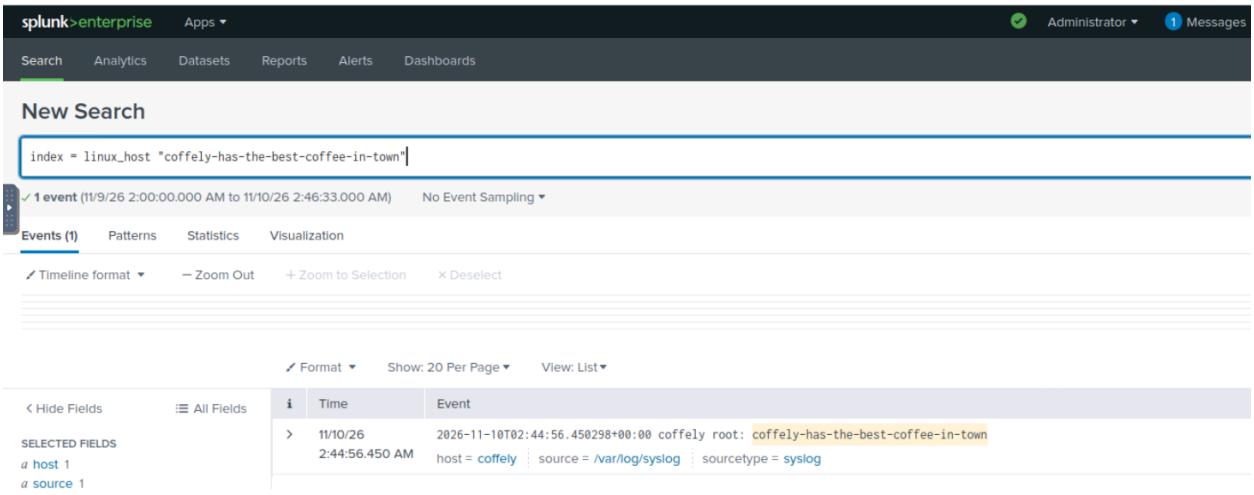
Screenshot 14

Syslog Monitoring Configuration



Screenshot 15

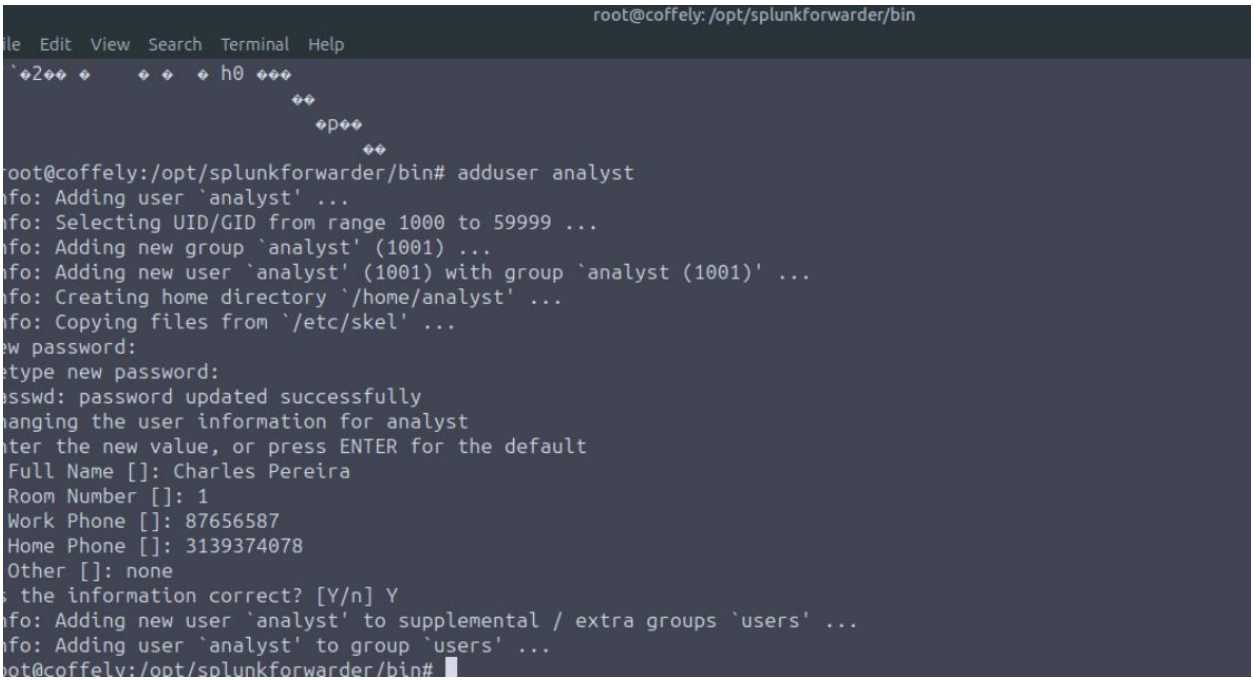
Data Ingestion Validation using SPL



Screenshot 16

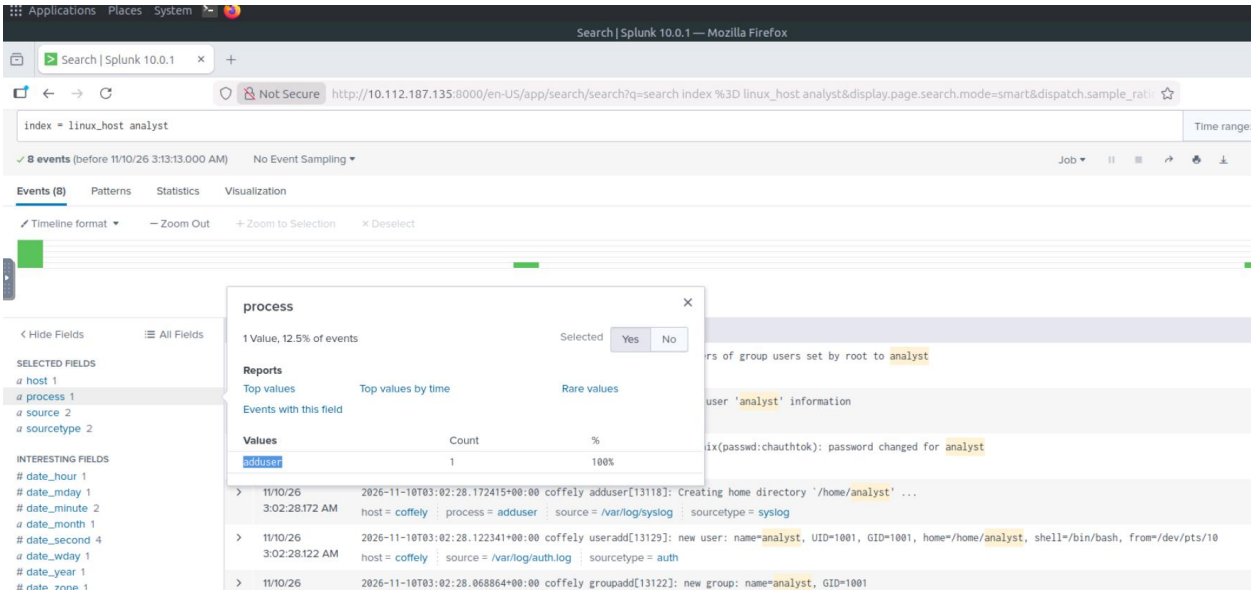
User Creation Event (adduser analyst)

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Screenshot 17

Authentication Monitoring Investigation



Project Outcome

The implementation successfully demonstrated the deployment of a fully functional SIEM environment capable of:

- Collecting Linux system logs
- Monitoring authentication events
- Aggregating infrastructure telemetry
- Centralizing event analysis
- Investigating user activities
- Performing SPL-based searches
- Supporting security operations workflows

Through Splunk Enterprise and Universal Forwarder integration, the environment provided visibility into operating system events and security-related activities, reproducing scenarios commonly found in Security Operations Centers (SOC).

Technical Environment

Platform

- Splunk Enterprise
- Splunk Universal Forwarder
- Ubuntu Linux

Data Sources

- /var/log/syslog
- /var/log/auth.log
- Apache Web Logs

Protocols

- TCP 9997
- Internal Splunk Management Services

Techniques Applied

- Log Forwarding
- Event Collection
- Indexing
- Monitoring
- Search & Reporting
- Security Event Investigation

Contact

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